GW Scattering Calculation Note

Name: **Lingyu Xia**
June 5, 2023

1 Separation of variables

Schrödinger's equation in flat space

$$\left(-\frac{1}{2M}\nabla^2 - \frac{N_A}{r_A} - \frac{N_B}{r_B}\right)\psi = \epsilon\psi \tag{1}$$

where ϵ is the energy of scattering state. Then we use the spheroidal coordinates with

$$\mu = \frac{r_A + r_B}{R}, \quad \lambda = \frac{r_A - r_B}{R}, \quad \varphi \tag{2}$$

As for the d'Alembertian applied on the scalar field ψ is in the form of $\Box\psi = \nabla^\mu\nabla_\mu\psi$. We first consider situation in flat spacetime, so the covariant operator is equal to partial operator

$$\Box\psi = \nabla^\mu\nabla_\mu\psi = \partial^\mu\partial_\mu\psi = g^{\mu\nu}\partial_\nu\partial_\mu\psi \tag{3}$$

Then we calculate the metric of spheroidal coordinate

$$r_A^2 = x^2 + (\frac{R}{2} + z)^2 \tag{4}$$

$$r_B^2 = x^2 + (\frac{R}{2} - z)^2 \tag{5}$$

$$\Longrightarrow \mu = \frac{1}{R}\left(\sqrt{x^2 + (\frac{R}{2} + z)^2} + \sqrt{x^2 + (\frac{R}{2} - z)^2}\right) \tag{6}$$

$$\lambda = \frac{1}{R}\left(\sqrt{x^2 + (\frac{R}{2} + z)^2} - \sqrt{x^2 + (\frac{R}{2} - z)^2}\right) \tag{7}$$

Then we calculate the scale factors

$$h_\mu = \frac{R}{2}\sqrt{\frac{\mu^2-\lambda^2}{\mu^2-1}} \tag{8}$$

$$h_\lambda = \frac{R}{2}\sqrt{\frac{\mu^2-\lambda^2}{1-\lambda^2}} \tag{9}$$

$$h_\phi = \frac{R}{2}\sqrt{(\mu^2-1)(1-\lambda^2)} \tag{10}$$

Then we have the Laplacian

$$\nabla^2\psi = \frac{1}{\sigma^2\left(\mu^2-\lambda^2\right)}\left\{\frac{\partial}{\partial\mu}\left[\left(\mu^2-1\right)\frac{\partial\psi}{\partial\mu}\right]+\frac{\partial}{\partial\lambda}\left[\left(1-\lambda^2\right)\frac{\partial\psi}{\partial\lambda}\right]\right\} \tag{11}$$

$$+\frac{1}{\sigma^2\left(\mu^2-1\right)\left(1-\lambda^2\right)}\frac{\partial^2\psi}{\partial\varphi^2} \tag{12}$$

where $\sigma = \frac{R}{2}$.

We introduce the Separation of variables, because ψ in independent of the ϕ, we separate the ψ in the form of

$$\psi = \sum_m a_m F_m(\mu) G_m(\lambda) e^{im\varphi} \tag{13}$$

Substitute it into the original equation

$$G_m\partial_\mu\left[(\mu^2-1)\partial_\mu\right]F_m + F_m\partial_\lambda\left[(1-\lambda^2)\partial_\lambda\right]G_m +$$
$$\left[-\frac{(\mu^2-\lambda^2)m^2}{(\mu^2-1)(1-\lambda^2)} + 2M\mu(N_A+N_B) - 2M\lambda(N_A-N_B) + 2M\epsilon\sigma^2(\mu^2-\lambda^2)\right]F_mG_m = 0$$

We separate the two variables in the form of $\mathrm{LHS}(\mu, F(\mu)) = \mathrm{RHS}(\lambda, G(\lambda)) = \rho_m$, then we have

$$\left[\frac{d}{d\mu}\left(1-\mu^2\right)\frac{d}{d\mu}+\frac{m^2}{1-\mu^2}-\rho_m+2M\sigma\left(N_A+N_B\right)\mu+2M\sigma^2\epsilon\mu^2\right]F_m(\mu)=0 \tag{14}$$

$$\left[\frac{d}{d\lambda}\left(1-\lambda^2\right)\frac{d}{d\lambda}-\frac{m^2}{1-\lambda^2}-\rho_m-2M\sigma\left(N_A-N_B\right)\lambda-2M\sigma^2\epsilon\lambda^2\right]G_m(\lambda)=0 \tag{15}$$

2 Power Series Solutions

We focus on the solutions near the z axis, where $\mu \sim 1$ and $\lambda \in (-1, 1)$. Then we could just consider Eq. . On the basis of the Leaver's Eq. 3

$$\frac{d}{d\mu}\left[\left(1-\lambda^2\right)\frac{dG_m}{d\lambda}\right]+\left[-\omega^2\lambda^2-2a\left(N_1-N_2\right)\lambda+A_{lm}-m^2/\left(1-\lambda^2\right)\right]G_m=0 \tag{16}$$

where $\omega = \sqrt{2M\sigma^2\epsilon}$, $a = M\sigma$, $A_{lm} = \rho_m$.

And then we should transform it into another form. Let $G_m(\lambda) = (1-\lambda^2)^{\frac{m}{2}}g(\lambda)$

$$(1-\lambda^2)g_{,\lambda\lambda} - 2(m+1)\lambda f_{,\lambda} + \left[\omega^2\lambda^2 + 2a(N_1-N_2)\lambda + m(m+1) - A_{lm}\right] g = 0 \tag{17}$$

On the basis of the work by leaver, we give its power series solutions. First, we should write the standard form for this differential equation. We let $x = \lambda + 1$

$$\begin{aligned} &x(x-2)g_{,xx} + 2(m+1)(x-1)g_{,x} + \\ &\left[\omega^2 x(x-2) + 2a\left(N_1-N_2\right)(x-2) \ + \omega^2 + 2a\left(N_1-N_2\right) + m(m+1) - A_{lm}\right] g = 0. \end{aligned} \tag{18}$$

$$x(x-x_0)\frac{d^2y}{dx^2} + (B_1+B_2x)\frac{dy}{dx} + \left[\omega^2 x(x-x_0) - 2\eta\omega(x-x_0) + B_3\right] y = 0 \tag{19}$$

Following Baber and Hasse, a power series solution about $x = 0$ may be obtained by letting

$$y(x) = e^{i\omega x}\sum_{n=0}^{\infty} a_n^\theta x^n \tag{20}$$

The sequence of expansion coefficients $\{a_n^\theta : n = 1, 2, \cdots\}$ is defined by the three-term recurrence relation

$$\begin{aligned} &\alpha_0^\theta a_1^\theta && +\beta_0^\theta a_0^\theta + && = 0, \\ &\alpha_n^\theta a_{n+1}^\theta && +\beta_n^\theta a_n^\theta + \gamma_n^\theta a_{n-1}^\theta && = 0, n = 1, 2, \end{aligned} \tag{21}$$

where

$$\begin{aligned} \alpha_n^\theta &= -x_0 n^2 + (B_1 - x_0)\, n + B_1, \\ \beta_n^\theta &= n^2 + (B_2 - 2i\omega x_0 - 1)\, n + 2\eta\omega x_0 + i\omega B_1 + B_3, \\ \gamma_n^\theta &= 2i\omega n + i\omega\left(B_2 - 2\right) - 2\eta\omega. \end{aligned} \tag{22}$$

We list the constants in the Eq.

$$\begin{aligned} &B_1 = 2(m+1), \quad B_2 = -2(m+1), \quad B_3 = \omega^2 + 2a(N_1-N_2) + m(m+1) - A_{lm} \\ &x_0 = 2, \quad \eta = \frac{2a(N_2-N_1)}{\omega}, \omega = \sqrt{2M\sigma^2\epsilon}, a = M\sigma, A_{lm} = \rho_m \end{aligned} \tag{23}$$

3 Numerical Calculation

3.1 Power Series Solutions

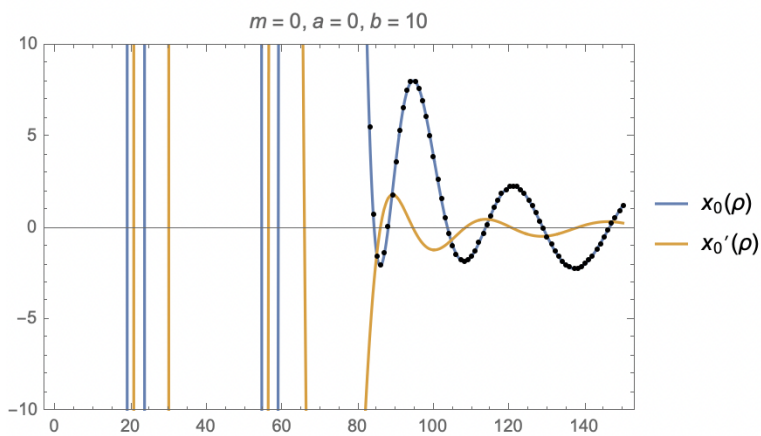
Then we use mathematica to calculate numerical solution. We set the M is mass of sun $M = 1.9891 \times 10^{30} kg$, R is 100 times the diameter of the solar system $R = 9 \times 10^{12} m$, ϵ is the energy of incident wave, which satisfies $\epsilon = h\nu = \nu \times 6.62 \times 10^{-34}$. Then we use following mathematica's code to plot this power series solution

```
ClearAll[\[Alpha], \[Beta], \[Gamma], F, A]
\[Alpha][n_] := -2 n^2 + (B1 - 2) n + B1
\[Beta][n_] :=
 n^2 + (B2 - 4 I \[Omega] - 1) + 4 \[Eta] \[Omega] + I \[Omega]
    B1 + B3
\[Gamma][n_] :=
 2 I \[Omega] n + I \[Omega] (B2 - 2) - 2 \[Eta] \[Omega]
A[0] = 1;
A[1] = -\[Beta][0]/\[Alpha][0];
Do[A[n] = (\[Beta][n - 1]*A[n - 1] + \[Gamma][n - 1]*
      A[n - 2])/\[Alpha][n - 1], {n, 2, range}]
F[x_] := Exp[\[Omega] x] Sum[x^n A[n], {n, 0, range}]
Plot[{Re[F[x]], Re[F'[x]]}, {x, 0, 0.01}]
```

Then, we can use this power series function to get initial condition for original spheroidal equation close to the point A numerically. We use mathematica's NDSolveValue function to solve spheroidal equation with power series solution's initial condition.

```
ClearAll[eigen]
清除全部
eigen[ρ_] := Module[{B1, B2, B3, ω, η, b, a, xcut, Eq, sol, y, x, m, ssol},
                 模块
  m = 0;
  a = 0;
  b = 10;
  B1 = -2 (m + 1);
  B2 = 2 (m + 1);
  B3 = -ρ + m (m + 1) + a;
  ω = b i;
  η = -a/(2 ω);
  xcut = 0.00001;
  Eq = x (x - 2) y''[x] + (B1 + B2 x) y'[x] + (ω^2 x (x - 2) + a (x - 2) + B3) y[x];
  sol = NDSolveValue[{Eq == 0, y[xcut] == 1, y'[xcut] == 41/2}, y, {x, xcut, 2 - xcut}];
        数值解的值
  sol[2 - xcut]]
```

Then we use Interpolation function to fit this solution and plot it in following figure.



To find the constants from separation variables, we should clarify the criterion that what condition should be satisfied for ρ. Obviously, the first derivative of λ should be zero at B point, then our criterion is that the first derivative is zero corresponding to the orange line in the above figure. Therefore, we just use function FindRoot to derive the roots.

4 Taylor Expansion

If we consider GW at point B, the effect of point A can be neglected. The partial wave can be expressed by:

$$\int d\rho a_m(\rho) F_m(\rho,\mu) G_m(\rho,\lambda) = \frac{1}{r_A}\sum_l b_{lm} H_l^{A+}(x_A)\sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\theta_A) \tag{24}$$

where

$$x_A = (\lambda+\mu)\sqrt{2M\epsilon}, \quad r_B = \sqrt{r_A^2 + R^2 - 2r_A R\cos\theta_A}, \quad \cos\theta_A = \frac{1+\lambda\mu}{\lambda+\mu}$$

For a certain partial wave, we have

$$\frac{2}{R(\mu+\lambda)} H_l^{A+}\left(\sqrt{2M\epsilon}\frac{R}{2}(\mu+\lambda)\right)\sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m\left(\frac{1+\lambda\mu}{\lambda+\mu}\right) = \int d\rho a_{lm}(\rho) F_m(\rho,\mu) G_m(\rho,\lambda) \tag{25}$$

For line segment $z \in (-\sigma,\sigma)$, $\mu = 1$ so the left-hand is nonzero only for $m \neq 0$

$$\frac{1}{\sigma(\mu+\lambda)}\mathcal{F}_0^A(\omega(\mu+\lambda)) = \int d\rho a_{l0}(\rho) F_0(\rho,1) G_0(\rho,\lambda) \tag{26}$$

where,

$$\mathcal{F}_0^A(x) = \frac{e^{-\pi\lambda_A/2}\left|\Gamma\left(l+1+i\lambda_A\right)\right|}{2(2l+1)!}(-i)^{l+1} M_{i\lambda_A, l+\frac{1}{2}}(2ix) = \frac{1}{2i}\left[H_l^{A+}(x) - H_l^{A-}(x)\right] \tag{27}$$

Then we expand both sides of equation, and align the term of $(1+\lambda)$ order by order. The Taylor expansion can be obtained in Leaver's paper. For the case of m = 0, one has,

$$G_0(\rho,\lambda) = e^{i\omega(\lambda+1)}\sum_{n=0}^{+\infty} c_n^\lambda(\rho)(\lambda+1)^n \tag{28}$$

$$\begin{aligned} &\alpha_0^\theta c_1^\theta && +\beta_0^\theta c_0^\theta + && = 0, \\ &\alpha_n^\theta c_{n+1}^\theta && +\beta_n^\theta c_n^\theta + \gamma_n^\theta c_{n-1}^\theta && = 0, n = 1,2,\cdots \end{aligned} \tag{29}$$

where

$$\begin{aligned} \alpha_n^\lambda &= -2n^2 - (B_2+2)\,n - B_2, \\ \beta_n^\lambda &= n^2 + (B_2 - 4i\omega - 1)\,n - a - i\omega B_2 + \rho + m(m+1), \\ \gamma_n^\lambda &= 2i\omega n + i\omega\,(B_2 - 2) + a, \end{aligned} \tag{30}$$

and,

$$\begin{aligned} a &= 2M\sigma\left(N_A - N_B\right) = -2\omega\left(\lambda_A - \lambda_B\right), \\ B_2 &= 2(m+1). \end{aligned} \tag{31}$$

Assume that $c_0^\theta = 1$, and we can calculate each term for c_λ^θ

$$\begin{aligned} c_1^\lambda &= \frac{\rho}{2} - \frac{1}{2}(a + 2i\omega) \\ c_2^\lambda &= \frac{1}{16}\rho^2 + \frac{1}{8}(1 - a - 4i\omega)\rho + \frac{1}{16}\left(a^2 + 8ia\omega - 12\omega^2\right) \\ c_3^\lambda &= \frac{1}{288}\rho^3 + \frac{1}{288}(8 - 3a - 18i\omega)\rho^2 + \frac{1}{288}\left(3a^2 + 36ia\omega - 6a - 92\omega^2 - 36i\omega + 12\right)\rho \\ &\quad - \frac{1}{288}\left(a^3 + 18ia^2\omega + 2a^2 - 92a\omega^2 - 120i\omega^3 + 8\omega^2\right) \\ &\quad \ldots \end{aligned} \tag{32}$$

Then we define "nth moment" as,

$$Z_n = \int d\rho a_{l0}(\rho) F_0(\rho, \mu)\rho^n \tag{33}$$

Thus, Eq. (26)

$$\begin{aligned} e^{i\omega(\lambda+1)} \Big\{ & Z_0 + \frac{1}{2}(\lambda+1)\left[-(a+2i\omega)Z_0 + Z_1\right] \\ & + \frac{1}{16}(\lambda+1)^2\left[\left(a^2 + 8ia\omega - 12\omega^2\right)Z_0 + 2(1 - a - 4i\omega)Z_1 + Z_2\right] \\ & + \frac{1}{288}(\lambda+1)^3\left[-\left(a^3 + 18ia^2\omega + 2a^2 - 92a\omega^2 - 120i\omega^3 + 8\omega^2\right)Z_0\right. \\ & \left. + \left(3a^2 + 36ia\omega - 6a - 92\omega^2 - 36i\omega + 12\right)Z_1 + (8 - 3a - 18i\omega)Z_2 + Z_3\right] \\ & + \mathcal{O}\left((\lambda+1)^4\right) \Big\} \end{aligned} \tag{34}$$

And we can use Mathematica to expand LHS of Eq.(26) to obtain each moment

5 NEW THOUGHTS WITH TAYLOR EXPANSION

For $m = 0$ wave at $\lambda \to -1$, we have,

$$\frac{1}{\sigma(\mu+\lambda)} H_0^{A+}(\omega(\mu+\lambda)) = \int d\rho a_0(\rho) F_0(\rho, \mu) G_0(\rho, \lambda)$$

[lx, hz] where $\omega = \sqrt{2M\epsilon}\sigma$ and $H_l^{A\pm}(x)$ is defined as,

$$\frac{\partial^2}{\partial x^2} H_l^{A\pm}(x) + \left[1 - \frac{l(l+1)}{x^2} - \frac{2\lambda_A}{x}\right] H_l^{A\pm}(x) = 0.$$

[lx,hz] Comparing to Eq. (1), one has,

$$x = \omega r_A/\sigma,$$
$$\lambda_A = -\sqrt{\frac{M}{2\epsilon}} N_A.$$

[lx] The $H_l^{A\pm}$ can be written in terms of Whittaker functions,

$$H_l^{A\pm}(x) = (\mp i)^l e^{(\pi\lambda_A/2)\pm i\sigma_l(\lambda_A)} W_{\mp i\lambda_A, l+\frac{1}{2}}(\mp 2ix),$$

[lx, hz] where,

$$\sigma_l(\lambda_A) = \arg\left(\Gamma\left(l+1+i\lambda_A\right)\right)$$

[lx,hz] The $\pm$sign labels the asymptotic behaviour of the solution,

$$H_l^{A\pm}(x) \to e^{\pm i\left[x-\lambda_A \log(2x)-\frac{l}{2}\pi+\sigma_l(\lambda_A)\right]}.$$

[lx,hz] However, Eq. (6) has a severe flaw. The LHS diverges at A. A more realistic calculation should be,

$$\frac{1}{\sigma(\mu+\lambda)}\mathcal{F}_0^A(\omega(\mu+\lambda)) = \int d\rho a_0(\rho) F_0(\rho,\mu) G_0(\rho,\lambda)$$